Introduction to Artificial Intelligence Assignment 3

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1 Explanation of Methods

1.1 Constructing the Bayesian Network

The network structure mirrors the physical layout of the hurricane problem. The root node is Weather, which directly influences the Flooding nodes for every edge in the graph. For the Evacuees nodes at each vertex, I implemented the Noisy-OR model as specified in the requirements. This model allows the network to calculate the probability of evacuees based on whether any of the incident edges are flooded, avoiding the need for exponentially large conditional probability tables for vertices with many connections.

1.2 Reasoning Algorithm

I used Inference by Enumeration to answer the probability queries. This standard algorithm sums over the hidden variables to calculate exact posterior probabilities. To determine if a specific path is free from flooding, I treated the edges in the path as "evidence." I momentarily set those edges to "not flooded" in the evidence list and calculated the conditional probability of that specific configuration occurring, given the existing observations.

2 Example Runs

2.1 Scenario 1: Basic Diamond Graph

Input Configuration:

```
#V 4          ; number of vertices n in graph (from 1 to n)
#P1 0.3       ; Value of parameter P1

#E1 1 3 W1 F 0.2  ; Edge1 between 1 and 3
#E2 2 3 W3 F 0.1  ; Edge2 between 2 and 3
#E3 2 4 W3 F 0.3  ; Edge3 between 2 and 4
#E4 3 4 W4 F 0    ; Edge4 between 3 and 4
```

```
#W 0.1 0.4 0.5 ; Prior distribution over weather
```

Execution Command:

```
python3 src/main.py -n --graph_path configs/scenario_1.txt \
        --q4_edges "1,4" \
        --q5_start 1 \
        --q5_end 4
```

Output:

```
--- Probabilistic Reasoning Report ---
1. What is the probability that each of the vertices contains
    evacuees?
  Vertex 1: 0.3360
  Vertex 2: 0.7133
  Vertex 3: 0.4739
  Vertex 4: 0.6480

2. What is the probability that each of the edges is flooded?
  Edge 1: 0.4800
  Edge 2: 0.2400
  Edge 3: 0.7200
  Edge 4: 0.0000

3. What is the distribution of the weather variable?
  P(mild) = 0.1000
  P(stormy) = 0.4000
  P(extreme) = 0.5000

4. Probability path ['1', '4'] is free from flooding:
  P(Free) = 0.5200

5. Path from 1 to 4 with highest probability of being free:
  Best Path (Edges): [1, 4]
  Probability: 0.5200
```

2.2 Scenario 2: The Bridge Graph

Input Configuration:

```
#V 6
#P1 0.6

#E1 1 2 W2 F 0.4
#E2 2 3 W2 F 0.4
#E3 3 6 W2 F 0.4

#E4 1 4 W5 F 0.1
#E5 4 5 W5 F 0.1
#E6 5 6 W5 F 0.1

#E7 2 5 W3 F 0.2

#W 0.2 0.6 0.2
```

Execution Command:

```
python3 src/main.py -n --graph_path configs/scenario_2.txt \
        --q4_edges "4,5,6" \
        --q5_start 1 \
        --q5_end 6
```

Output:

```
1. What is the probability that each of the vertices contains
    evacuees?
  Vertex 1: 0.6070
  Vertex 2: 0.8246
  Vertex 3: 0.7622
  Vertex 4: 0.3179
  Vertex 5: 0.5269
  Vertex 6: 0.6070

2. What is the probability that each of the edges is flooded?
  Edge 1: 0.7600
  Edge 2: 0.7600
  Edge 3: 0.7600
  Edge 4: 0.2000
  Edge 5: 0.2000
  Edge 6: 0.2000
  Edge 7: 0.4000

3. What is the distribution of the weather variable?
  P(mild) = 0.2000
  P(stormy) = 0.6000
  P(extreme) = 0.2000

4. Probability path ['4', '5', '6'] is free from flooding:
  P(Free) = 0.5216

5. Path from 1 to 6 with highest probability of being free:
  Best Path (Edges): [4, 5, 6]
  Probability: 0.5216
```